PRASUN KUMAR

AI/ML Technical Lead | GenAI Solution Architect | Agentic AI & RAG Specialist

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PROFESSIONAL SUMMARY

AI/ML Technical Lead and GenAI Solution Architect with 9+ years of full-time software engineering experience since July 2017. Designs and delivers enterprise AI platforms across Agentic AI, Large Language Model (LLM) applications, Retrieval-Augmented Generation (RAG), document intelligence, AI automation, automated testing, and cloud-native integration. Strong focus on architecture ownership, stakeholder alignment, production readiness, event-driven design, human-in-the-loop controls, idempotency, retries, dead-letter queues (DLQs), privacy controls, observability, and cost-aware LLM engineering.

CORE SKILLS

AI & GenAI: LLMs, GPT, Llama, Gemma, Qwen, Mistral, IBM Granite, prompt engineering, structured extraction

Agentic AI: LangGraph, LangChain, multi-agent workflows, tool calling, function calling, memory, approval gates

RAG: vector search, semantic search, hybrid search, embeddings, chunking, metadata filtering, retrieval evaluation

ML / DL: scikit-learn, XGBoost, NLP, classification, recommendation systems, feature engineering, PyTorch, TensorFlow, Hugging Face

MLOps / LLMOps: MLflow, Docker, Kubernetes, CI/CD, model registry, experiment tracking, evaluation, monitoring

Platform / Cloud: Azure OpenAI, AI Foundry patterns, Azure Functions, Service Bus, Blob Storage, Azure AI Search, Document Intelligence, FastAPI, PostgreSQL, MySQL

Leadership: solution architecture, delivery planning, design reviews, stakeholder workshops, mentoring, production readiness, governance

PROFESSIONAL EXPERIENCE

Associate Technical Lead / Technical Lead, AI and ML | Harbinger Group Aug 2024 - Present

Lead architecture and delivery for enterprise GenAI, Agentic AI, RAG, document intelligence, AI automation, and AI evaluation initiatives across Azure-based workflows.

Architected HAS platform capabilities spanning Discovery Agent, Automation Tester, and EvalAI Pro: multi-tenant SaaS UI/API, Azure Functions, Service Bus, Blob Storage, PostgreSQL, Azure OpenAI, Azure AI Search, and Jira/Azure DevOps integrations.

Architected a GenAI email-to-case automation platform using Azure Functions, Service Bus, Blob Storage, Azure OpenAI, OCR/document intelligence, MySQL, Graph API, and case-management integrations.

Designed an AI-driven Automation Tester pipeline that converts Jira/ADO work items into LLM-generated test cases and Playwright API/UI scripts, executes them, and routes results/defects to Xray/ADO/Jira.

Improved document extraction accuracy by 60%+ through better OCR normalization, prompt/schema design, confidence scoring, validation, and human review loops.

Designed RAG patterns with chunking, semantic/hybrid retrieval, metadata filtering, citation support, and quality evaluation to improve grounded knowledge discovery.

Led behavior scoring and text classification work using NLP/LLM patterns, including Llama/Gemma/Qwen experimentation, QLoRA fine-tuning concepts, adapter lifecycle practices, and evaluation dashboards.

Reduced model/storage footprint by 77 TB through lifecycle cleanup, governance, artifact review, and storage optimization practices.

Mentor engineers, lead design reviews, manage stakeholder discussions, and define production-readiness checks for AI delivery teams.

Senior Software Engineer | Harbinger Group Apr 2021 - Jul 2024

Delivered enterprise software and AI/ML-adjacent systems across backend services, automation workflows, integrations, document processing, and data-driven applications.

Built and integrated APIs, data-processing flows, automation utilities, and reporting components with emphasis on maintainability and reliability.

Collaborated with product, QA, DevOps, and client stakeholders to convert business requirements into scalable technical implementations.

Improved engineering delivery through code reviews, reusable patterns, troubleshooting support, and mentoring of junior engineers.

Software Professional | Extentia Information Technology Nov 2019 - Apr 2021

Developed backend, ETL, and integration solutions using Python, pandas, REST APIs, database workflows, and cloud-aligned delivery patterns.

Implemented NLP chatbot patterns using intent recognition, entity extraction, dialogue flow, BERT-style embeddings, and Elasticsearch search workflows.

Built AI/automation prototypes including resume parsing and ATS matching, translation API patterns, and computer-vision based attendance workflows.

Associate IT Applications Specialist | Symantec Software India Pvt Ltd Jul 2017 - Oct 2019

Supported enterprise application engineering, automation, data workflows, and operational reliability for internal business systems.

Built Python automation for database patching, refreshes, MySQL backups, SOX audit reporting, and LMS JSON-template conversion.

Created Flask/MySQL dashboards for ServiceNow tickets and server-performance visibility, improving operational productivity by 30%.

Master Level Intern | Symantec Software India Pvt Ltd Jan 2017 - Jul 2017

Worked on enterprise application support and automation foundations before moving into full-time engineering.

SELECTED AI ARCHITECTURE PROJECTS

GenAI-Powered Email-to-Case Automation Platform (Technical Lead / AI Solution Architect): Email ingestion, OCR, LLM extraction, automated case creation, Azure OpenAI, Azure Functions, Service Bus, Blob Storage, MySQL, retry handling, idempotency, audit trail, and human validation.

Enterprise RAG Knowledge Platform (RAG Architect): Document ingestion, chunking, embeddings, hybrid retrieval, Azure AI Search patterns, retrieval APIs, citations, confidence thresholds, and evaluation metrics.

HAS Multi-Tenant Azure GenAI Platform (AI Platform Architect / Technical Lead): Discovery documents to redacted RAG index, multi-agent analysis, BRD/GAP generation, Jira/ADO work-item sync, tenant-scoped config, retries, observability, and production-parity local debugging.

AI-Driven Automation Tester Platform (AI Automation Architect): Jira/ADO story intake, LLM test-case generation, Playwright API/UI script generation, Service Bus handoffs, Blob report storage, Xray/ADO result sync, and Jira Bug creation through existing reporting flows.

Agentic AI Multi-Agent Platform (Agentic AI Specialist): Planner, research, retrieval, validator, executor, tool routing, human approval, retry, memory, and audit-trace patterns using LangGraph-style orchestration.

AI Behavioral Scoring Platform (Technical Lead): NLP/text classification experimentation using BERT/RoBERTa/SentenceTransformers/XGBoost-style patterns, multilingual support considerations, threshold calibration, and evaluation practices.

EvalAI Pro and LLM Evaluation (LLMOps / Evaluation): Agent evaluation patterns using RAGAS, DeepEval, MLflow-style tracking, run metrics, drift analysis, and artifact persistence for release-readiness decisions.

PUBLIC GITHUB PORTFOLIO

enterprise-ai-ml-case-studies - sanitized enterprise GenAI/RAG/document AI/agentic workflow case studies with runnable POCs

agentic-ai-langgraph-workflows - multi-agent workflow with approval gates, retries, tools, and audit traces

enterprise-rag-quality-lab - RAG chunking, retrieval metrics, grounding, citation coverage, and evaluation examples

document-intelligence-llm-pipeline - OCR + LLM extraction pipeline with redaction, schema validation, confidence scoring, and review routing

email-to-case-genai-automation - event-driven GenAI automation with idempotency, retries, DLQ, validation, and audit logs

ai-resume-job-matcher - explainable resume/JD matching with skill extraction, weighted scoring, gaps, and responsible AI notes

production-llm-platform-on-kubernetes - FastAPI LLM gateway, prompt registry, policy checks, metrics, Docker, and Kubernetes manifests

privacy-preserving-rag-dlp - privacy layer for RAG with PII/secret detection, redaction, policy decisions, citations, tests, Docker, and CI

ai-jailguard / ai-soc-alert-copilot - defensive LLM security and SOC triage demos with APIs, CLI workflows, tests, Docker, and CI

EDUCATION AND CERTIFICATION

Post Graduate Program in Machine Learning & Artificial Intelligence - Credential: credential.net/20453200-981b-4985-9ddf-2a44c5ee66cf

Master of Computer Applications (MCA), Artificial Intelligence - Pune University, 2017.

Bachelor of Computer Applications (BCA), Computer Applications - Sikkim Manipal University, 2014.

AWARDS AND RECOGNITION

3x Quarterly Team Award Winner; 2x Superstar Award Winner; 4x Technical Star Recognition.

Additional recognitions include We-NOT-Me, going-the-extra-mile, Thank You, FunStar, PEP Silver, and Hackathon Recognition - 2017.

Recognized between 2017 and 2026 for technical excellence, innovation, leadership, ownership, collaboration, and delivery impact.

THOUGHT LEADERSHIP

Shares practical perspectives on Agentic AI, AI agents in production, GenAI architecture, RAG systems, LLM reliability, AI governance, and emerging AI trends.

YouTube: youtube.com/@AIWizardry277